Term Test, Second Semester 2020

MATHS 350 SC: Topology

(Time allowed: 50 MINUTES)

1. Let $\{X_\alpha : \alpha \in A\}$ be a collection of topological spaces.

(a) (3 marks) Give a basis for the Tychonoff topology (product topology) on $X = \Pi_{\alpha\in A} X_\alpha$.

(b) (7 marks) Show that the Tychonoff topology is the weakest topology on $X = \Pi_{\alpha\in A} X_\alpha$ for which each of the projections π_α is continuous.
[Show that if σ is a topology on X and τ is the Tychonoff topology on X, then $\tau \subseteq \sigma$.]

2. Suppose X is a topological space and $A \subseteq X$.

(a) (5 marks) Prove that if there is a sequence of points in A converging to x, then $x \in \overline{A}$.

(b) (5 marks) Prove that if X is metrisable, then the converse of (a) also holds.

3. (a) (5 marks) Suppose that $f, g : X \to Y$ are continuous functions and Y is Hausdorff. Prove that the set $\{x \in X : f(x) = g(x)\}$ is closed.

(b) (5 marks) Prove that $\mathbb{R}$ (with the usual topology) is regular.

4. (a) (4 marks) Prove that if X is second countable then X is separable and first countable.

(b) (3 marks) Give an example of a space that is first countable but not separable.

(c) (3 marks) Give an example of a space that is separable but not first countable.